



Solving two step linear equations

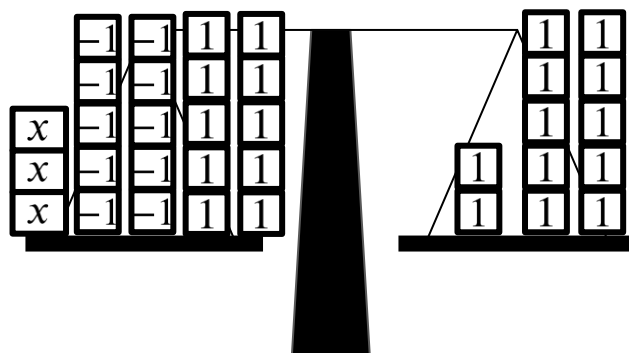
Task A: Solving using an additive step first

$$3x - 10 = 2$$

$$3x - 10 + 10 = 2 + 10$$

$$3x = 12$$

$$x = 4$$



2) Solve the following equations:

a) $9x + 7 = 25$

$$9x + 7 + (-7) = 25 + (-7)$$

$$9x = 18$$

$$x = 2$$

b) $9x - 7 = 25$

$$9x - 7 + 7 = 25 + 7$$

$$9x = 32$$

$$x = \frac{32}{9}$$

c) $4 = 5y + 19$

$$4 + (-19) = 5y + 19 + (-19)$$

$$-15 = 5y$$

$$y = -3$$

d) $\frac{1}{2}x + 7 = 25$

$$\frac{1}{2}x + 7 + (-7) = 25 + (-7)$$

$$\frac{1}{2}x = 18$$

$$x = 36$$

Task B: Multiple approaches to solving the same equation

1) Solve these equations two different ways.

a) $2x + 18 = 30$

$$\frac{2x + 18}{2} = \frac{30}{2}$$

$$x + 9 = 15$$

$$x = 6$$

b) $3d - 27 = 9$

$$\frac{1}{3}(3d - 27) = \frac{1}{3}(9)$$

$$d - 9 = 3$$

$$d = 12$$

c) $35 - 7x = 21$

$$\frac{35 - 7x}{7} = \frac{21}{7}$$

$$5 - x = 3$$

$$x = 2$$

$$2x + 18 = 30$$

$$3d - 27 = 9$$

$$35 - 7x = 21$$

$$2x + 18 + (-18) = 30 + (-18) \quad 3d - 27 + 27 = 9 + 27 \quad 35 - 7x + (-35) = 21 + (-35)$$

$$2x = 12$$

$$3d = 36$$

$$-7x = -14$$

$$x = 6$$

$$d = 12$$

$$x = 2$$

2) Solve: $\frac{13}{4} - \frac{x}{8} = \frac{7}{2}$

$$8\left(\frac{13}{4} - \frac{x}{8}\right) = 8\left(\frac{7}{2}\right)$$

$$26 - x = 28$$

$$x = -2$$